Ben McKallip

Round Rock, TX | (512) 983-8611 | mckallipb02@gmail.com
GitHub: @bmck039 | LinkedIn: linkedin.com/in/ben-mckallip | Website: bmck039.github.io

EDUCATION

Southwestern University, Georgetown, TX (*August 2022 – May 2026*)
Bachelor of Science in Physics and Computer Science, Minor in Mathematics

- Graduated *Cum Laude* with a 3.780 / 4.00 GPA across 165 credit hours
- Dean's List: Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, and Spring 2025

HONORS AND AWARDS

- **Feagin Scholarship** (Elizabeth Feagin Brown and Foy F. Feagin, Jr. Scholarship), Southwestern University - merit scholarship awarded to outstanding physics majors
- **Cody Scholar Award**, Southwestern University - top-tier merit scholarship for academic excellence
- **King Creativity Award**, Southwestern University (*Fall 2024 - Spring 2025, Fall 2025 - Spring 2026*)
- **Upsilon Pi Epsilon** Honor Society (Computer Science)
- **Pi Mu Epsilon** Honor Society (Mathematics)
- **Omicron Delta Kappa** Honor Society (Leadership)
- Dean's List, six semesters
- Cum Laude graduate

RESEARCH EXPERIENCE

Gravitational Wave Research (*Summer 2025 – Current*)
University of Texas at Dallas, NSF Research Experiences for Undergraduates (REU)

- Worked independently in collaboration with graduate students and a Principal Investigator on questions at the intersection of gravitational lensing and binary black hole precession
- Used computational and analytical methods to differentiate the effects of regular precession and gravitational lensing on binary black hole inspiral waveforms
- Systematically varied chirp mass, time delay, flux ratio, precession amplitude and frequency, and phase parameters to identify regions of the parameter space where the two effects become degenerate
- Discovered notable degeneracy between precession and lensing parameters, and identified regions where the two can be reliably distinguished
- Estimated the prevalence and detectability of gravitational microlenses given current detector signal-to-noise constraints
- Worked extensively with PyCBC and matched filtering techniques, running high-performance computing workloads on TACC Lonestar
- Contributed a bug fix upstream to PyCBC (see Open-Source Contributions under Projects)

Artificial Intelligence Undergraduate Research (*Spring 2024*)
Southwestern University, Department of Computer Science

- Applied adversarial-training algorithms in creating intelligent card-game agents using Python
- Designed agents to reason under imperfect information and coordinate with a partner in the game of Spades
- Benchmarked three approaches: Expectiminimax with probabilistic bidding, Monte Carlo Tree Search, and a deep Reinforcement Learning agent trained through self-play using the SIMPLE library inside Docker
- Worked in collaboration with 3 other students, achieving intermediate human-level play with consistent partner coordination

PUBLICATIONS

In preparation:

- Nguyen Vo, T. N., Singh, T., McKallip, B., Kesden, M., and King, L. *Distinguishing lensing and precessional modulation in binary black-hole inspiral waveforms*.

PROJECTS

Full-Stack Computer Science Capstone (*Spring 2026*)
Go • Svelte • PostgreSQL • GCP • Cloudflare • Linux

- Developed a full-stack platform matching 1,600 students with 70 campus organizations at Southwestern University
- Wrote the backend, database schema, authentication system, and REST API using Go and PostgreSQL
- Set up automated transactional email delivery through Google Cloud Platform
- Configured reverse-proxy production hosting using a Cloudflare Tunnel
- Provisioned a Linux virtual machine as a shared development server with remote SSH access via Cloudflare for the team
- Configured the Svelte build pipeline and supported 3 teammates on frontend integration and the matching algorithm
- Handed off to the Office of Student Life; official adoption pending

Fish Tank Controller - IoT Automation System (*Personal Project*)
Python • Flask • BLE • JavaScript • Machine Learning

- Built a full-stack IoT platform for automated aquarium management
- Designed a browser-based dashboard for real-time monitoring and control of tank hardware
- Wrote a Python BLE backend and a Flask web server, all coordinated by a single startup script
- Trained a machine-learning model on live sensor data to monitor tank conditions and flag anomalies

Open-Source Contribution to PyCBC (gwastro/pycbc PR #5146**)** (*2025*)
Python • SciPy • Matched filtering • pytest

- Traced sharp discontinuities in PyCBC's `optimized_match()` function, discovered while running mismatch calculations for gravitational wave research
- Root cause: an unbounded `scipy.optimize.minimize_scalar` call escaping the valid time-shift window
- Switched to the bounded optimizer and added a regression test with HDF5 fixture data
- Pull request approved by PyCBC core maintainers

Astrus Chat Application (github.com/bmckallip/Astrus**)** (*2015 – 2024*)
WebSockets • Custom authentication • Full-stack

- Wrote a full-stack chat application with WebSocket-based real-time messaging
- Designed and implemented a custom secure login system
- Long-running personal project started in 8th grade and iterated on through early college; served as a self-taught foundation for later formal computer science work

Multi-Agent Spades AI (*Spring 2024*)
Python • PyTorch • Docker • Reinforcement Learning

- Built a framework implementing the card game Spades and benchmarking three AI strategies
- Applied Expectiminimax with probabilistic bidding, Monte Carlo Tree Search, and deep Reinforcement Learning algorithms in creating intelligent card-game agents using Python
- Trained the RL agent through self-play in Docker using the SIMPLE library
- Recorded win rates and score distributions across simulated tournaments and visualized results automatically
- Achieved intermediate human-level play through partner coordination under imperfect information

Quadruped Robot Library (*Fall 2024 - Spring 2025*)
C++ • Arduino • Inverse kinematics • Embedded systems

- Wrote a C++ Arduino library for a custom-built quadruped robot with 12 Tower Pro servo motors controlled through a breakout board and an Arduino Nano
- Implemented inverse kinematics with per-leg and per-joint movement functions
- Wrote serial-based calibration methods and a templated time-execution helper for smooth multi-joint motion
- Built in collaboration with Collin Cook under faculty advising as part of the King Creativity Fund

ESP32 Smart-Home IR Controller (*Summer 2026*)
ESP32 • ESPHome (YAML) • Home Assistant (Python) • IR (custom RC5)

- Built an ESP32-based IR receiver/emitter integrated with Home Assistant via ESPHome
- Enabled multi-device macro control: one remote press triggers multiple commands across devices
- Wrote a custom RC5 instruction set to work around hardware limitations of the IR receiver module

TEACHING EXPERIENCE

Physics Tutor, Southwestern University (*Fall 2023 – Spring 2024, Fall 2025 – Spring 2026*)

- Tutored undergraduate students enrolled in Physics I and Physics II
- Guided students through core concepts, homework problems, and exam preparation
- Adapted explanations to individual learning styles and levels of preparation

TECHNICAL EXPERIENCE

IT Student Technician, Southwestern University (*Spring 2024 – May 2026*)

- Provided hardware and software technical support to faculty and students
- Maintained campus IT equipment including workstations, printers, and network infrastructure
- Handled imaging workflows and general equipment lifecycle management
- Automated software installation and initial setup for campus computers

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, Go, C, C++, Java, C#, PHP, HTML, CSS, SQL, (familiar with Haskell, Prolog, Lisp, Curry)
Web and Backend: Svelte / SvelteKit, Node.js, Express, Gorilla Mux, REST API design, WebSockets, OAuth, Nginx
Databases: PostgreSQL, MongoDB, Neo4j, SQL schema design
AI and Machine Learning: PyTorch, TensorFlow, NumPy, SciPy, scikit-learn, LLM APIs (OpenAI, Anthropic), prompt engineering, Reinforcement Learning, Monte Carlo Tree Search, Expectiminimax
Cloud and DevOps: Docker, Google Cloud Platform, Cloudflare (Tunnels, DNS), GitHub Actions, CI/CD, Linux server administration, Bash scripting
Embedded and Hardware: Arduino, ESP32, ESPHome, Raspberry Pi, BLE / Bluetooth, IR protocols (custom RC5), inverse kinematics, basic circuit design (resistor networks, transistors, motors, passive components), 3D printing
Physics and Scientific Computing: PyCBC, matched filtering, high-performance computing, Grace plotting tool, DS9 image analysis, Rebound n-body simulation
Tools and Systems: Git, LaTeX, Home Assistant, Adobe Creative Cloud

RELEVANT COURSEWORK

Computer Science: Artificial Intelligence, Database Management, Senior Seminar in Software Engineering, Algorithms, Discrete Mathematics

Mathematics: Differential Equations, Vector Calculus, Signals and Systems, Linear Algebra, Complex Analysis, Introductory Statistics
Physics: General Relativity, Undergraduate Astrophysics, Quantum Mechanics, Math Methods in the Physical Sciences, Electromagnetism, Electronics

ACTIVITIES AND LEADERSHIP

- **Physics Club** - Officer, Southwestern University
- **Computer Science Club** - Member, Southwestern University